

Quality & supplier-audit, the way your team actually wants to work.

An AI-native EQMS + supplier-audit platform built to GxP rigor. Compliant by construction, configurable to your process, and designed so your QA team spends less time on paperwork and more on quality.

Technical & User Briefing · April 2026

WHAT YOU GET

One platform, 15 modules, the digital thread you've been asked to build.

INTERNAL QUALITY — 8 CLOSED-LOOP MODULES

Document control, deviation/NC, CAPA, change control, training, complaint/OOS, risk management, management review. Every record e-signed, audit-trailed, ALCOA+ compliant from day one.

SUPPLIER AUDIT — FULL LIFECYCLE

Risk classification, qualification, pre-audit prep, on-site/remote execution, findings, supplier CAPA, periodic re-audit, for-cause triggers. Public-data fusion built in.

CROSS-MODULE WIRING

A deviation auto-creates a CAPA. A document change auto-assigns retraining. A failed CAPA triggers a for-cause audit. Audit/event signals feed supplier scorecards. The digital thread, working.

AI THAT DRAFTS; HUMANS APPROVE

20+ AI agents handle the slow work — audit prep, finding drafts, report assembly, supplier risk dossiers. Every output cites its source. The AI never signs — a qualified human always does.

LIVE PUBLIC-DATA FUSION

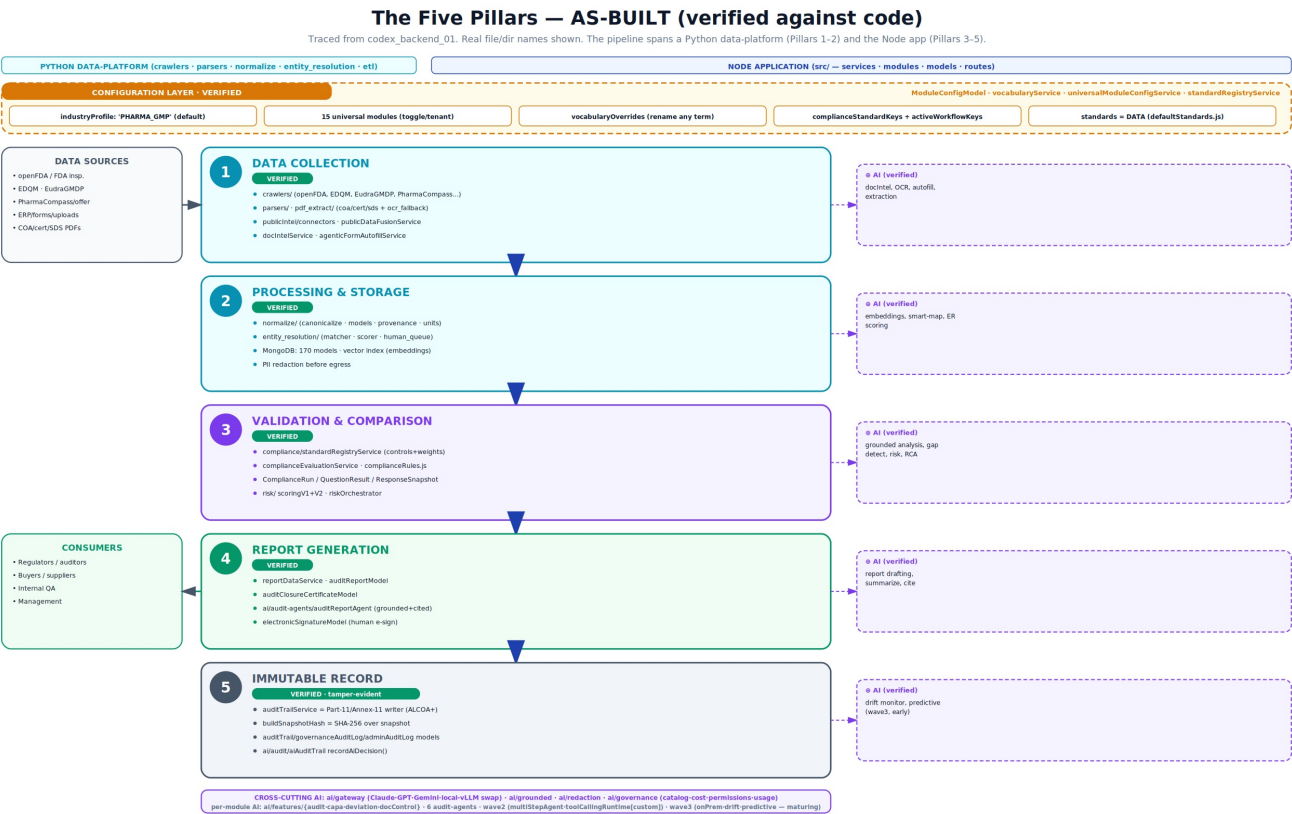
FDA inspections, EMA EudraGMDP, WHO data stream continuously into supplier risk profiles. You see what the regulator already knows, automatically.

DEPLOYMENT, YOUR CHOICE

Cloud, hybrid, or on-prem. Your data residency, your sovereignty requirements, your security model.

HOW IT'S BUILT

Five pillars: collect, process, validate, report, seal — with AI plugged in at every step.



VERDICT: all 5 pillars exist as discrete, real components. Pillars 1-2 = Python data-platform; 3-5 = Node app. Standards & vocabulary are DATA (config), confirming engine/config separation.
Honest nuance: AI attaches per-module & per-wave (not literally one slot per pillar); validation today is audit/compliance-centric; on-prem is validation-gated not yet proven a2e; toolCallingRuntime is custom (not MCP).

Verified against the codebase — not a marketing diagram. The configuration layer (top) is what makes the engine industry-agnostic; the AI gateway swaps models without changing the pipeline. Standards (ICH Q7, 21 CFR, GMP) are data, not code.

Built to the standards your inspector will check.

Requirement	How Hawkeye satisfies it
21 CFR Part 11 / Annex 11 e-signatures	Re-authentication at the moment of signing; binds signer + meaning + timestamp + record version. Signed records are locked against edit.
ALCOA+ data integrity	Every record Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available. Enforced structurally, not optionally.
Append-only audit trail	Every CRUD action, transition, and signature logged with before/after state, actor, reason. Human-readable, searchable, exportable. Cannot be modified.
Tamper-evident records	Per-record SHA-256 hashing of signed snapshots. Any post-signature modification is detectable.
Separation of duties	System administrator role distinct from quality approver. Role overlaps are flagged and auditable.
Validated state (IQ/OQ/PQ)	Vendor-provided IQ/OQ documentation; PQ supported on your data. Clear delineation of vendor vs. customer validation responsibility.
AI governance (ISO 42001-aligned)	Per-AI-decision reproducibility record: inputs, prompt, model, sources, confidence. Inspector-defensible.

AI drafts. Humans decide. Every output is reproducible.

PER-CALL REPRODUCIBILITY For every AI output, we record: the inputs retrieved, the prompt version, the model used, the confidence score, and the redactions applied. An inspector can ask "how was this generated?" and we show the full trace.	PER-TENANT AI CONTROLS Enable/disable each AI agent per module. Choose the model (Anthropic, OpenAI, Gemini, or your own on-prem model) per task. Set grounding/confidence thresholds. Govern access by RBAC.
GROUNDED-OR-FALLBACK Every output cites its source. If the AI cannot ground a claim in retrievable data, it returns "insufficient evidence" rather than guessing. No hallucinated regulatory advice.	PII REDACTION BEFORE EGRESS Sensitive identifiers are stripped before any external model sees them. Configurable PII classes. Defence-in-depth.
HUMAN-IN-THE-LOOP, ENFORCED AI never commits, approves, or signs a record. A qualified human reviews and e-signs every output. This is enforced at the platform level — it cannot be configured away.	WORKS WITH AI OFF The platform is a deterministic workflow + records system. Turn AI off and it still operates as a full EQMS. AI is leverage, not dependency.

From request to closed, with AI assistance at every step.

Statuses and the transitions between them, with the gate (who/what) on each transition.



Representative lifecycle. Confirm exact status names and transition rules against the live workflow engine

Findings linked to CAPAs. Supplier scorecard updated. All artifacts retained per your data-retention policy.

HOW IT FITS IN YOUR STACK

Open APIs. Standard connectors. The digital thread without re-platforming.

WHAT WE CONNECT TO

ERP: SAP, Oracle — supplier master sync.

LIMS/MES: standard APIs for batch + test data.

Document repository: SharePoint, controlled DMS.

SSO: Azure AD, Okta, SAML, OAuth.

Notification: email, Teams, Slack.

WHAT WE INGEST

Structured: ERP master data, LIMS results, ticketing systems.

Unstructured: PDFs (COA, CoC, SDS), scanned audit reports, emails.

Public: FDA inspections, EMA EudraGMDP, WHO PQ data, recall lists.

OPEN APIS (READ & WRITE)

Every major object — audits, CAPAs, deviations, documents, suppliers — exposed via documented, versioned REST APIs. Event-driven webhooks for near-real-time integrations. Custom-connector pattern documented.

VALIDATION-COMPATIBLE

Integrations are part of the validated state. Change control on connector configuration. API access governed by RBAC + audit-trailed. Your CSV documentation includes them.

Enterprise-grade, configurable to your policy.

Control	How it works
Authentication	SSO (SAML, OAuth, Azure AD, Okta) + MFA enforced. Named accounts only — no shared logins for any action that can be signed.
RBAC with configurable roles	Per-capability permissions. Roles can be scoped to specific sites, suppliers, or modules. Standard roles (Viewer, Auditor, QA Approver, Admin) are cloneable starting points.
Separation of duties	System Admin cannot approve quality records by default. Role overlaps are explicit and audit-logged. Change to a user's permissions is itself audited.
Session management	Configurable session timeout, lockout, re-authentication policy. Re-auth required at every e-signature.
Data encryption	TLS 1.3 in transit; AES-256 at rest. Per-tenant data isolation in multi-tenant deployments.
Data residency	Configurable region. India, EU, US, on-prem options supported. Critical for export-quality pharma.
Backup & DR	Configurable RPO/RTO appropriate to GxP records. Disaster recovery documented and testable.

Configurable without code — six surfaces, one console.

Admin Configuration Map — Audit Management

The full admin surface: what an administrator can configure, grouped by area. Representative — verify against the live Admin console.

Users & Access

Users: invite, deactivate, SSO map

Roles: define & assign (RBAC)

Permission matrix per role

Approver / signer designation

Session & 2FA policy

Audit Configuration

Audit types (supplier/internal/for-cause)

Standards library (WHO-GMP, 21 CFR...)

Questionnaire templates

Severity & classification schemes

Numbering / ID format

Workflow & Lifecycle

Status flow & transitions

Approval steps (single/multi)

Mandatory-field rules

"Findings must link CAPA" gate

Escalation & due-date rules

AI Governance

Enable/disable agents per type

Model selection via gateway

Grounding threshold & fallback

PII redaction policy

On-prem / BYO-model endpoint

AI usage budget / credit caps

E-Signature & Compliance

Signature meanings list

Re-auth on signing (PIN/2FA)

21 CFR Part 11 settings

Record-locking policy

Data retention rules

Integrations & Data

Supplier master sync (ERP/SAP)

Public-data fusion sources

Document repository link

Export / API keys

Notification channels (email)

USERS & ACCESS

Roles, RBAC, SSO mapping, signer designation, MFA policy.

WORKFLOWS & LIFECYCLE

Status flows, transition rules, approval steps, mandatory fields, escalation triggers.

AI GOVERNANCE

Per-agent enable/disable, model selection, grounding threshold, PII policy, usage budget.

Configuration changes are themselves audit-logged. In a validated environment, treat the admin console as a controlled surface and route changes through your change-control process.

HAWKEYE

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WHAT CHANGES FOR THE PEOPLE WHO USE IT

Less paperwork. More quality work. Auditor-grade evidence, automatically.

QA ANALYST — THE DAILY USER

Before: 3-5 days per audit prep, gathering evidence manually, cross-referencing SOPs, building questionnaires from templates.

After: AI assembles the prep package in hours; analyst reviews and approves. The slow work becomes the review work.

LEAD AUDITOR — ON-SITE EXECUTION

Before: Paper checklists, photos on phone, late-night report writing.

After: Tablet-based questionnaire with autofill from evidence uploads; report draft ready before leaving the site.

QA APPROVER / HEAD

Before: Bottleneck on every audit close-out, every CAPA approval, every change control.

After: Approvals queued in priority order, cross-module context one click away, decisions in minutes not hours.

DOCUMENT CONTROLLER

Before: Manual training assignment on every SOP revision; chasing acknowledgements.

After: Training auto-assigned to affected roles on document revision; compliance dashboard tracks completion.

SUPPLIER QUALITY ENGINEER

Before: Self-reported supplier data, point-in-time risk assessment, surprise findings.

After: Continuous supplier risk profile fused with FDA/EMA/WHO public data. Early-warning alerts when supplier risk shifts.

IT / CSV TEAM

Before: Validation-heavy systems with manual data integration, multiple tools to maintain.

After: One validated platform, documented APIs, change-controlled configuration. Your CSV burden goes down, not up.

NEXT STEPS

See it on your own data.

A live sandbox with your audits, your suppliers, your SOPs. No commercial pressure. Full module access. Use it to evaluate, train your team, or run a parallel-run pilot alongside your current process.

WEEK 1

Sandbox provisioned. Your data loaded. Initial training session for 5-10 users.

WEEKS 2-4

Run one real audit end-to-end in parallel with your current process. Compare results.

WEEK 5+

Decide. If yes, transition planning. If not, keep the sandbox — it stays for your team to use.

Questions, technical deep-dives, integration architecture — [\[contact@hawkeye.app\]](mailto:contact@hawkeye.app)